



■ FX[®]-105HS

Carthage Mills' FX-105HS is a multipurpose nonwoven geotextile made of polypropylene staple fibers which are formed into a random network, needlepunched and heatset for dimensional stability. FX-105HS is part of the Carthage [FX-HS Series](#) of nonwoven geotextiles, is inert to biological degradation and resistant to naturally encountered chemicals, alkalis, and acids.

PROPERTY	TEST METHOD	DATA	
		METRIC	ENGLISH
☐ Mechanical			
Grab Tensile Strength	ASTM D 4632	1.20 kN	270 lbs
Grab Tensile Elongation		50%	
Trapezoidal Tear	ASTM D 4533	0.45 kN	100 lbs
CBR Puncture	ASTM D 6241	3.12 kN	700 lbs
☐ Endurance			
UV Resistance	ASTM D 4355	70% @ 500 hrs	
☐ Hydraulics / Filtration			
Permittivity ⁽¹⁾	ASTM D 4491	0.80 sec ⁻¹	
Water Flow Rate ⁽¹⁾		3055 lpm/m ²	75 gpm/ft ²
Apparent Opening Size (AOS) ^(1,2)	ASTM D 4751	0.150 mm	100 US Std. Sieve
☐ Physical			
Mass Per Unit Area (Typical)	ASTM D 5261	355.95 g/m ²	10.5 oz/yd ²
Standard Roll Sizes / Packaging / Weight	Measured (Typical)	3.81 m x 109.7 m 418 m ² 154 kg	12.5 ft x 360 ft 500 yd ² 340 lbs
		4.57 m x 91.5 m 418 m ² 154 kg	15.0 ft x 300 ft 500 yd ² 340 lbs

NOTES: Mullen Burst Strength ASTM D 3786 is no longer recognized by ASTM D35 on Geosynthetics. Puncture Strength ASTM D 4833 is not recognized by AASHTO M 288 and has been replaced with CBR Puncture ASTM D 6241.

- (1) At the time of manufacturing. Handling, storage and shipping may change these properties.
(2) Maximum Average Roll Value
- Unless otherwise stated, all values stated here are Minimum Average Roll Values (MARV).
 - The properties reported above are effective 01-01-26 and are subject to change without notice.

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